

Internship PROJECT

Palestine Tourism Recommendation System

Team Members:

- Asala Sadat - AI&ML (Team Lead)
- Thimar Arda - AI&ML
- Nadeen Shadeed - AI&ML
- Shahd Omari - Backend
- Daniah Zaheda - Backend
- Tala Sayara - Cybersecurity
- Tala Odeh - Frontend
- Deema Theeb - Mobile Application



A better way to discover Palestine

The challenge

Visitors need personalized recommendations based on their preferences—not a generic list of tourist attractions.

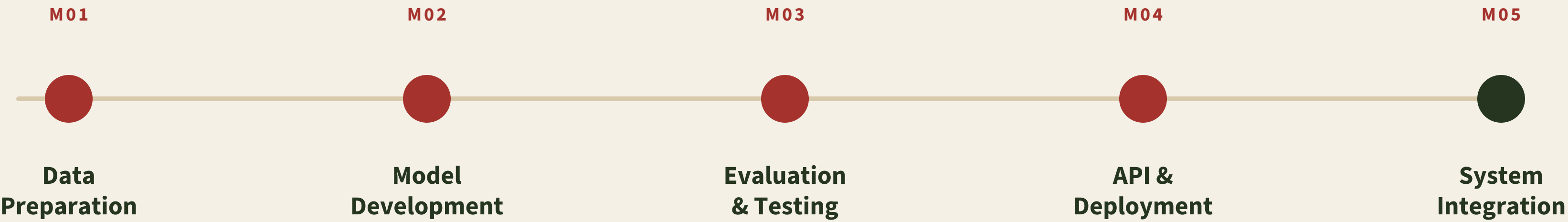
WHY IT MATTERS

Make every visit more personalized while promoting Palestine's diverse local destinations and communities.

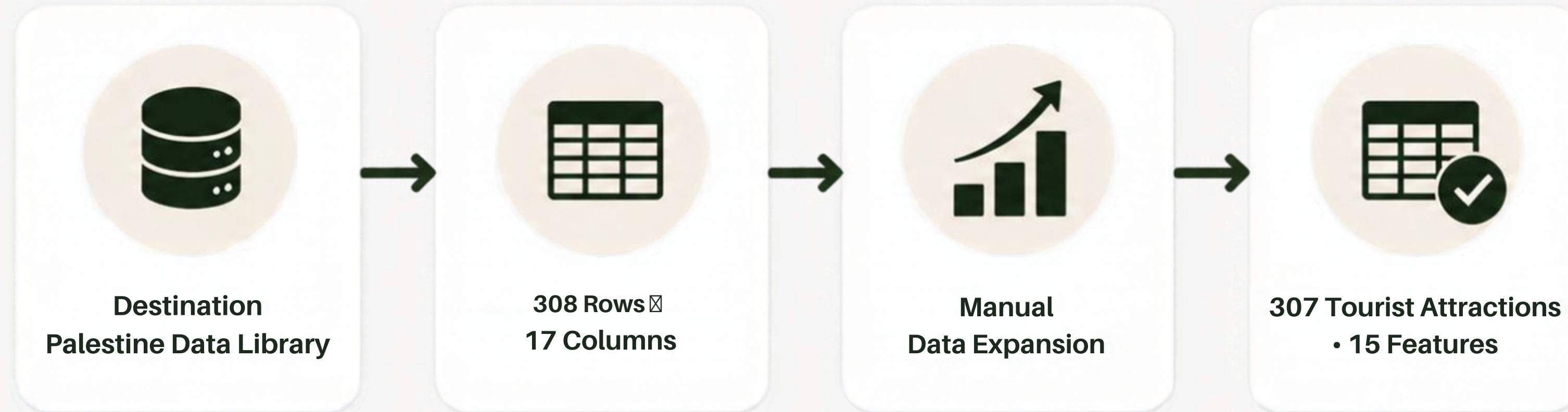
Accurate • User-friendly • Secure • Integrated



Project milestones



Dataset Collection & Enhancement



CHALLENGE:

The original dataset was too limited for our recommendation system.



SOLUTION:

We manually added and enriched tourism data with additional features such as:

City

Trip Type

Age Group

Budget

Description

Coordinates

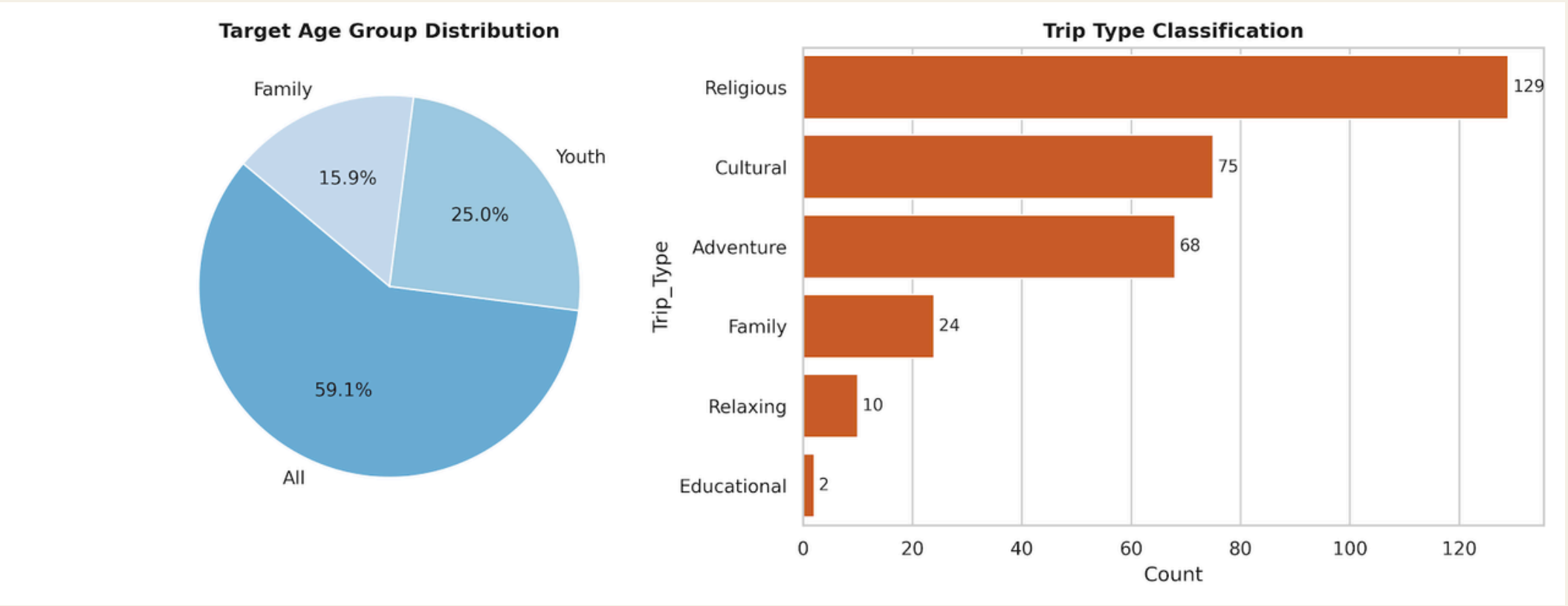
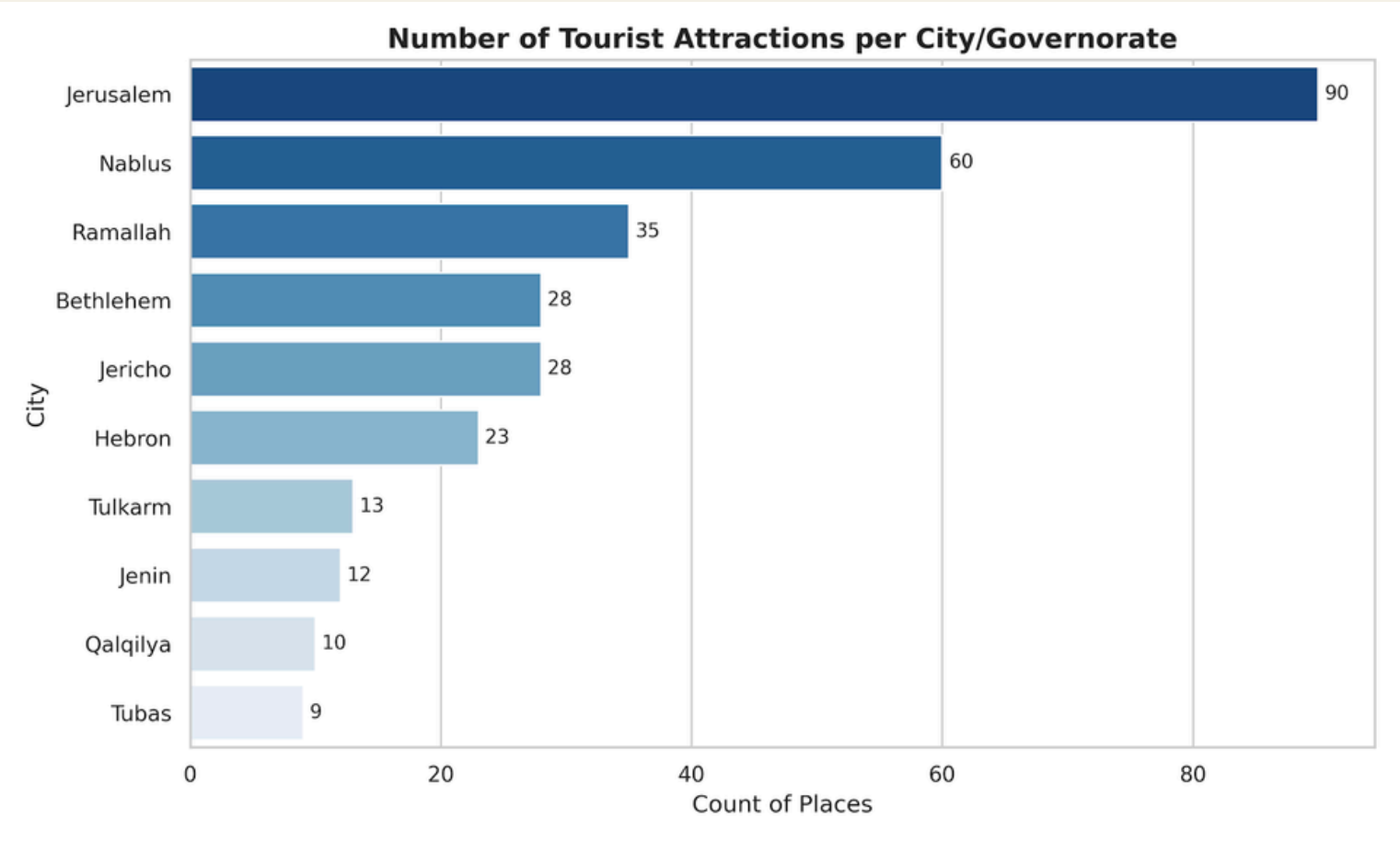
Image_URL

All 17 Dataset Columns & Specifications

308 Non-Null Rows (100%)

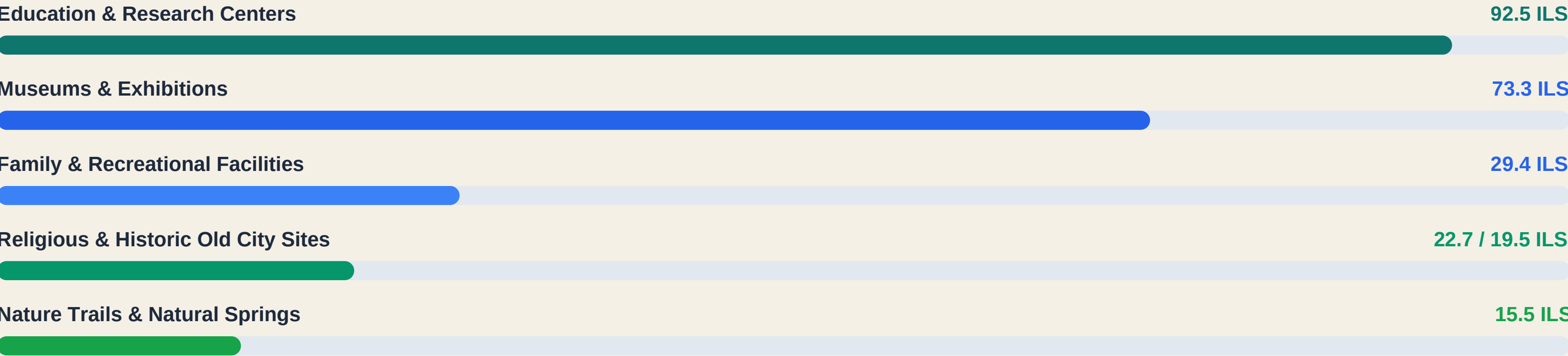
Column Name	Type	Non-Null	Unique	Description & Sample Values
ID	int64	308	308	Unique location identifier (1 to 308)
Place_Name	object	308	308	English title e.g. Mosque of 'Umar Ibn al-Khattab
type	object	308	9	Category: Religious, Historic Old City, Nature & Trails...
City	object	308	10	Governorate: Jerusalem, Nablus, Ramallah, Bethlehem...
Budget_Level	object	308	3	Tier classification: Low, Medium, High
Age_Group	object	308	3	Audience suitability: All, Youth, Family
Trip_Type	object	308	6	Activity type: Religious, Cultural, Adventure, Family...
Estimated_Cost_ILS	int64	308	22	Numeric cost value in Shekels (0 to 145 ILS)
Latitude / Longitude	float64	308	42	Geospatial coordinates (e.g. Lat 32.1905, Lon 34.9660)
Text & Media Fields	object	308	304+	Place_Name_AR, Description_AR, Description_EN, Image_URL

Exploratory Data Analysis



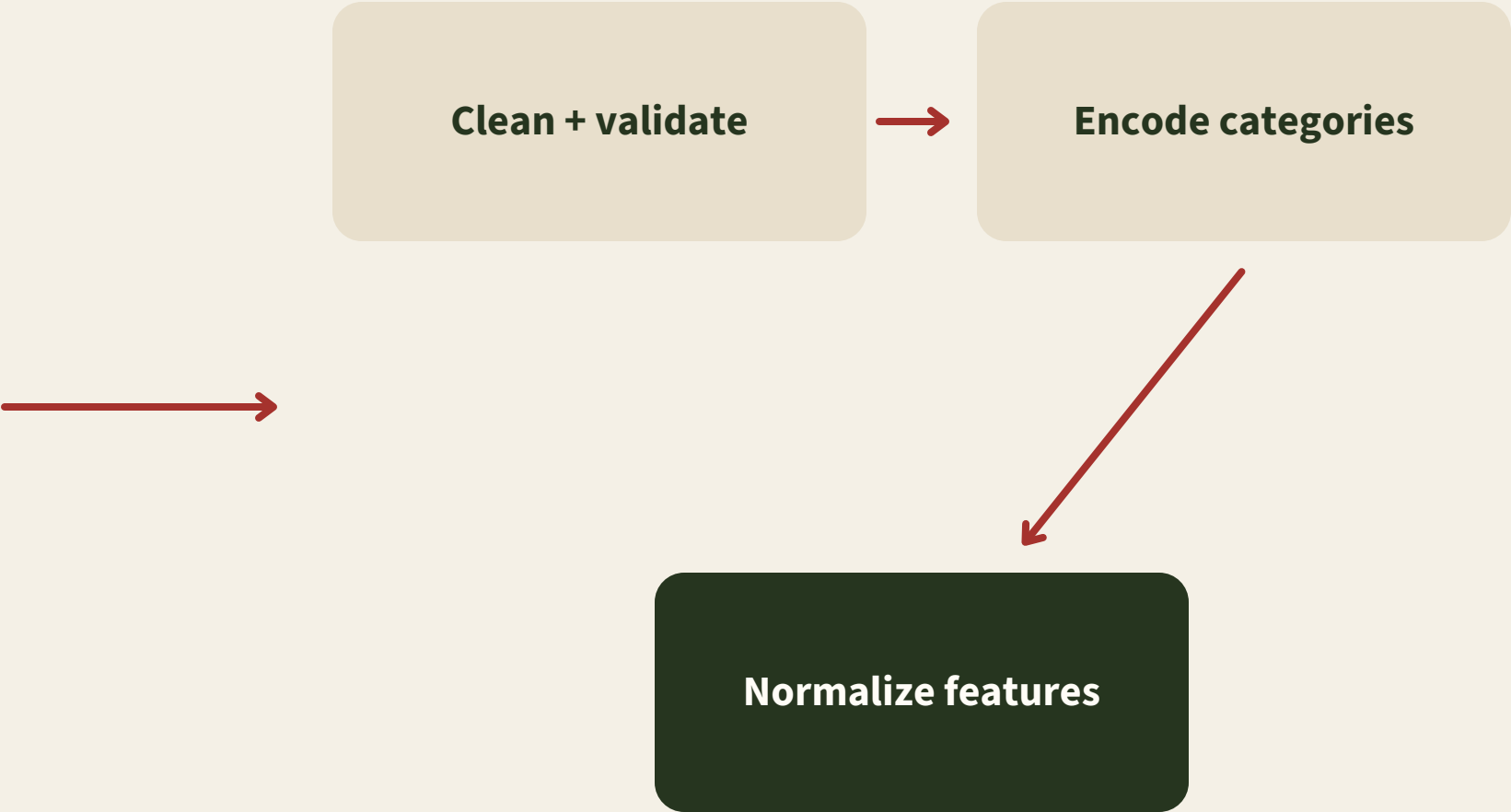
Exploratory Data Analysis

Average Estimated Cost by Tour's Type

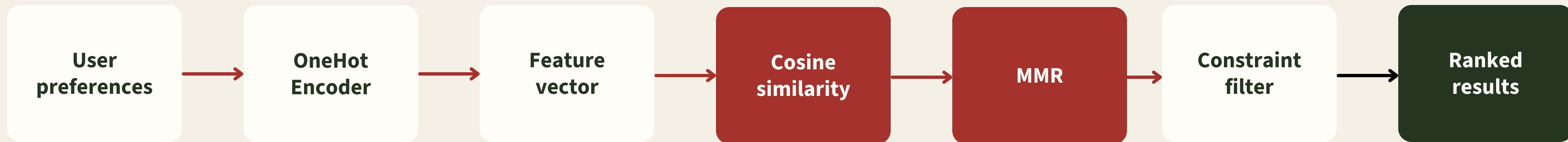


Turning tourism data into model-ready signals

CITY	TRIP TYPE	BUDGET	GROUP
Nablus	Culture	Medium	2
Jericho	Nature	Low	4
Bethlehem	History	High	3

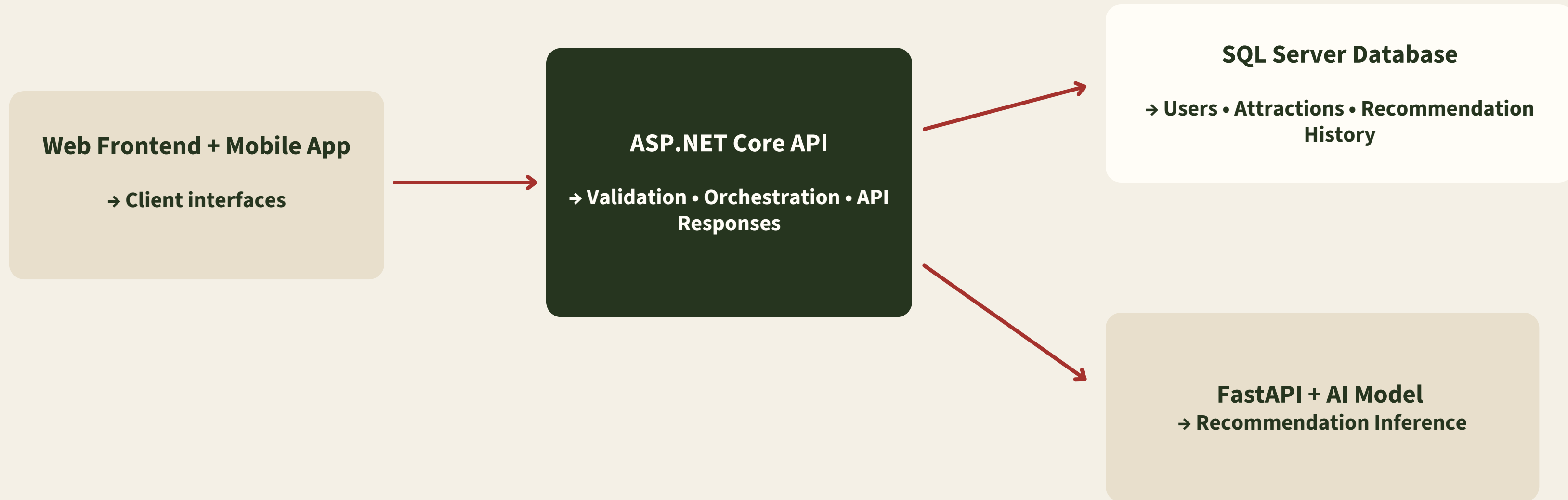


A clear, explainable recommendation pipeline

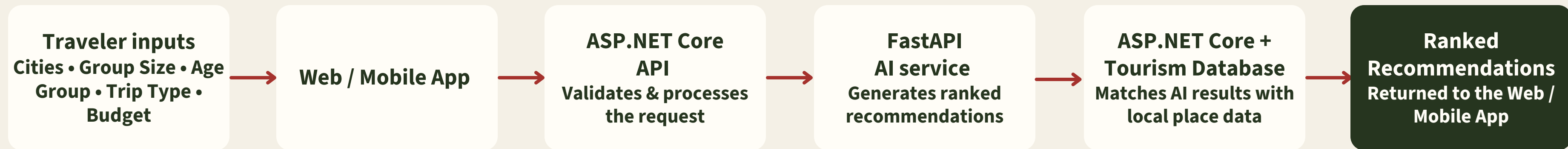


Relevant first. Feasible second. Ranked for the traveler.

Services with Clean Boundaries



From Traveler Preferences to Personalized Recommendations



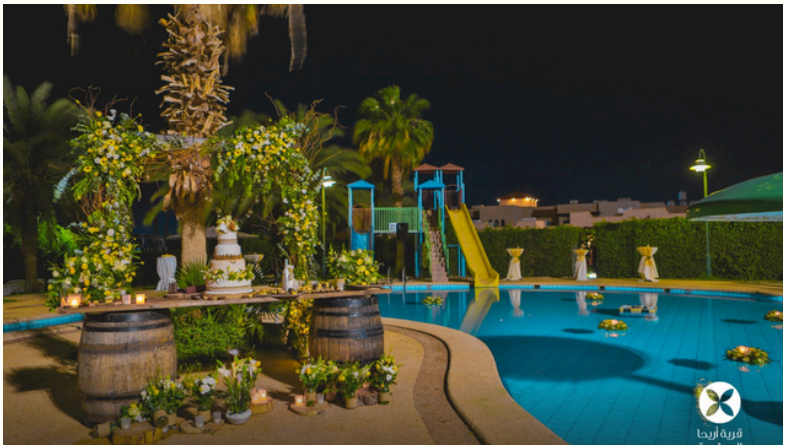
Preferences become a ranked, relevant shortlist in one seamless request.

Choose less. Discover more.

Plan your visit

CITY Jericho
TRIP History & Culture
BUDGET 500
GROUP 3 people
AGE GROUP Family

Recommended for you

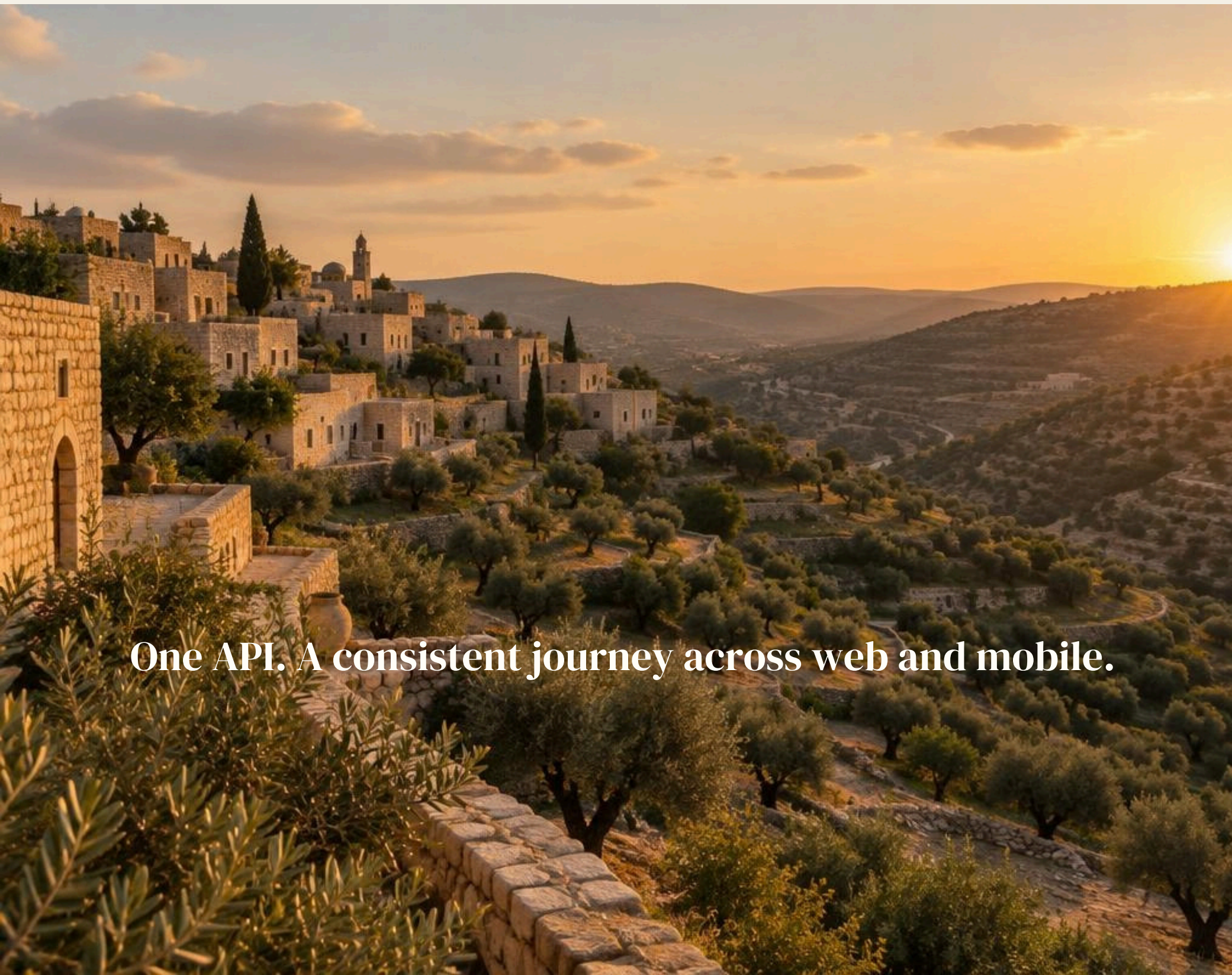


Jericho Tourist Park



Jericho Agro-Tourism Farms

Recommendations that travel with you



One API. A consistent journey across web and mobile.

Explore nearby

Jericho

Hisham's Palace
History • Nearby

Wadi trail
Nature • Half day

Trust is part of the architecture

**Authenticate
& validate**

Secure API access and strict request validation.

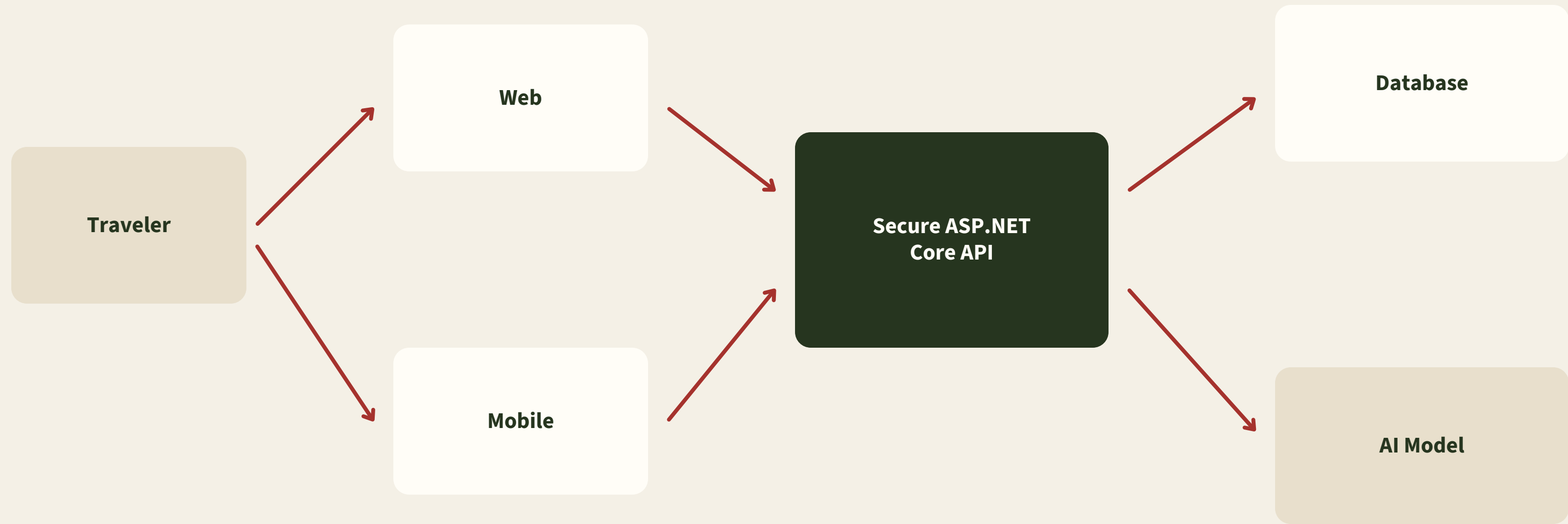
**Encrypt
in transit + at rest**

TLS connections and protected data storage.

**Protect privacy
by design**

Collect only what the experience needs.

One connected tourism platform



LIVE DEMO

Experience the recommendation system

Real-time preferences → AI-powered attraction suggestions

Personal discovery. Local impact.

A secure, multi-platform AI recommender built for Palestine’s destinations and travelers.

01
Personalized

02
Integrated

03
Secure

NEXT

Expand data • improve ranking • enrich features

Thank you for your attention!

Questions & Discussion

Palestine Tourism Recommendation System • Project Team

